

A Virtual-Reality Digital Twin Operator Display for Dynamic UAV Scene Reconstruction from In-Flight Detections

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Remote UAV operation requires the pilot to maintain spatial awareness of aircraft state, terrain, obstacles, and mission-relevant objects when direct visual contact or continuous high-quality video may be unavailable. This manuscript presents Pipeline B, a virtual-reality digital-twin operator display in which a pilot wearing a head-mounted display inspects a dynamically reconstructed 3D representation of the operational scene. The implemented runtime path converts in-flight object detections into georeferenced semantic telemetry and uses those data to spawn, update, mark stale, and remove dynamic actors inside an Unreal Engine and Cesium-based environment. The contribution is framed as a human-in-the-loop auxiliary operator display, not as autonomous flight control, certified detect-and-avoid, or a replacement for regulatory safety channels. The paper defines the runtime contract, entity lifecycle, georeferencing assumptions, VR interaction requirements, safety envelope, claim-to-evidence map, and validation protocol required for a complete VRIH submission. Quantitative results are retained as explicit placeholders until validation experiments, VR demonstration material, geospatial ground truth, network-degradation trials, and human-operator studies are completed.

Virtual reality; UAV; Digital twin; Scene reconstruction; Semantic telemetry; Teleoperation; Situational awareness

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1. Draft Status, Journal Target, and Placeholder Convention

This manuscript is a controlled working draft for *Virtual Reality & Intelligent Hardware* (VRIH). The intended article type is a full research paper, not a rapid communication. The reason is evidentiary: the argument requires an architecture, an implemented runtime contract, an immersive display concept, software instrumentation, quantitative results, and a human-operator validation. VRIH rapid communications are limited to four pages, whereas this work needs a complete method and validation package.

This version is not yet a submission-ready experimental article. It is deliberately honest: every unmeasured result, missing figure, unconfirmed hardware detail, statistical test, declaration, or physical-trial outcome is marked with [TBD: identifier and required evidence]. No quantitative claim should be treated as demonstrated until the corresponding placeholder has been replaced by measured evidence.

The placeholder convention used throughout the paper is:

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- **TBD-DEMO**: screenshots, HMD captures, video material, configuration records, and execution logs showing the implemented system operating in the VR display.
- **TBD-BW**: semantic telemetry bitrate, p95 bandwidth, packet rate, and comparison with H.264, H.265, or WebRTC video baselines.
- **TBD-LAT**: latency from detection publication to Brain ingest, Unreal actor update, and visible headset refresh.
- **TBD-GEO**: 2D and 3D reconstruction error against surveyed, RTK/GNSS, motion-capture, or otherwise defensible ground truth.
- **TBD-LOSS**: behavior under controlled packet loss, jitter, delay, and link interruption.
- **TBD-TRACK**: identity switches, stale-object duration, despawn correctness, and track fragmentation.
- **TBD-HUMAN**: pilot task success, near misses, workload, spatial awareness, cybersickness, and overtrust measures.
- **TBD-SAFETY**: operating limits, fallback rules, confidence thresholds, uncertainty thresholds, and stale-state visualization policy.
- **TBD-LOWVIS**: sensor-specific validation under night, smoke, fog, haze, glare, or equivalent degraded-visibility conditions.
- **TBD-FIG**: missing figures, HMD screenshots, video stills, diagrams, or graphical abstract assets.
- **TBD-STAT**: sample size, statistical model, effect size, confidence interval, and correction for multiple comparisons.
- **TBD-ETHICS**: ethics review or exemption for human-operator experiments, participant consent, and data protection.
- **TBD-REPRO**: dataset, code, log, configuration, and software-version release package.

2. Introduction

Remote UAV operation is a spatial decision-making problem. A pilot or mission operator must reason about aircraft pose, terrain, hazards, people, vehicles, mission targets, visibility, and communication degradation while often relying on a video stream and a map display. Video provides rich evidence, but it also binds the operator to a camera viewpoint, consumes bandwidth, degrades under compression, and can fail to show objects outside the current field of view. Map displays provide geospatial context, but they usually lack the immediacy and embodied spatial interpretation of a three-dimensional scene. The central problem is therefore not only communication load. It is the pilot's need to maintain situation awareness from partial, delayed, and heterogeneous evidence.

Virtual reality and telepresence research provide the conceptual basis for this operator-display problem. Virtual reality can be understood through the dimensions that determine telepresence, including sensory richness and vividness [1], while telepresence research distinguishes mediated presence from direct physical presence [2]. In dynamic systems, situation awareness includes perception of relevant elements, comprehension of their meaning, and projection of their future status [3]. Its measurement is nontrivial and must be designed explicitly rather than inferred from subjective impressions alone [4, 5]. Workload and human-factor measures such as NASA-TLX, SAGAT, SART, and simulator-sickness questionnaires are therefore part of the evidence burden for a VR operator display [6-9].

Pipeline B addresses this problem by converting detected objects into a persistent, spatially organized digital-twin layer. A detection source sends object-class, confidence, identity, timestamp, and estimated position data to a ground-station service called Brain. Brain normalizes and publishes the live entity layer, and Unreal Engine renders the resulting actors over Cesium geospatial context for an operator wearing a head-mounted display. The reconstructed scene is not presented as ground truth. It is an auxiliary operator display in which every dynamic actor must carry freshness, confidence, source, and uncertainty semantics.

The main research question is:

Can a VR-equipped UAV operator maintain better spatial awareness of mission-relevant objects under constrained or degraded communication than with video-only, map-only, or non-immersive hybrid displays, while preserving explicit visibility of stale, uncertain, or missing information?

This question cannot be answered by architecture alone. It requires a system demonstration, bandwidth measurements, latency measurements, geospatial error validation, degradation tests, and a human-operator study. This paper therefore separates what has already been implemented or specified from the physical and human evidence still required.

The intended contributions are:

- a VR digital-twin operator-display architecture for dynamic UAV scene reconstruction from in-flight detections;
- a concrete runtime contract linking detection ingestion, Brain state publication, Unreal/Cesium actor lifecycle, and HMD visualization;
- a formalized entity model, georeferencing model, latency model, bandwidth model, stale-state policy, and uncertainty-display requirement;
- a claim-to-evidence map that prevents architecture claims from being confused with validated experimental results;
- a VRIH-oriented validation protocol covering bandwidth, latency, geospatial accuracy, packet loss, tracking stability, human utility, low-visibility limits, safety envelope, and reproducibility.

3. Related Work and Positioning

Pipeline B sits at the intersection of immersive teleoperation, digital twins, semantic/task-oriented communication, UAV perception, UAV geolocation, and degraded-link human-computer interaction. Its novelty is not the invention of any one of those areas. The novelty claim must be narrower: an implemented and measurable runtime path for reconstructing UAV mission-relevant entities inside a VR digital twin from in-flight detections.

3.1 Immersive Teleoperation and Situation Awareness

Immersive interfaces have been studied as a way to improve operator perception in unmanned-vehicle control. Immersive telepresence with mixed reality has been evaluated for unmanned vehicle operation and situation awareness [10]. Multi-robot interface studies show that immersion and predictive display elements can affect operator situation awareness [11]. Recent work on VR teleoperation has also explored visual augmentation of live streams for unmanned vehicles [12] and compared 3D visualization and VR display designs for remote monitoring tasks [13]. These studies support the premise that immersion may change operator awareness, but they also show that the effect must be measured rather than assumed.

UAV-specific VR and digital-twin studies are directly relevant. Augmented-reality mobile digital twins have been proposed for UAV wildfire-prevention workflows [14]. VR-based teleoperation has been applied to UAV-manipulator systems [15]. Recent VR digital-twin work on UAV control studies how different points of view affect user experience and performance [16]. Pipeline B differs from these works by focusing on dynamic reconstruction from semantic detections rather than only viewpoint control, simulated piloting, or manipulation interfaces.

3.2 Digital Twins for VR and UAV Systems

Digital twin research has established terminology, design implications, and system taxonomies. VRIH has explicitly highlighted the connection between virtual reality, intelligent hardware, and digital twins [17]; VRIH has also published digital-twin work in applied AI contexts [18]. Broader surveys characterize digital twins as cyber-physical representations requiring data synchronization, modeling, and operational purpose [19-21]. For UAVs, digital twins have been discussed in beyond-5G contexts [22] and UAV-swarm cooperation [23]. These works motivate digital

twins as data-driven operational interfaces, but they do not by themselves validate a specific UAV-to-HMD operator workflow.

3.3 Semantic and Task-Oriented Communication

Semantic communication and task-oriented communication shift the objective from reconstructing every bit of a source signal to transmitting information useful for a downstream task. Surveys and early systems describe semantic communication as a future wireless-network paradigm [24], while deep-learning semantic communication systems demonstrate end-to-end learned communication mechanisms [25]. Task-oriented edge inference uses information bottleneck ideas to trade communication cost against task performance [26]. Edge-video analytics extends the idea to video-derived features rather than full video reconstruction [27], and cooperative edge inference studies multi-device task-oriented communication [28].

Pipeline B should not claim to solve semantic communication theory. Its object stream is closer to engineered semantic telemetry: it sends detected entities and their display-relevant attributes instead of a full video stream. The correct claim is operational and measurable: for the selected operator task, entity telemetry may reduce bandwidth while preserving relevant spatial information. That claim remains [TBD: TBD-BW bandwidth evidence and task-completeness evidence].

3.4 UAV Detection, Geolocation, and Communication Constraints

UAV object detection is mature enough that this paper should not frame detector choice as its primary novelty. Real-time object detection based on UAV remote sensing has been extensively reviewed, including hardware, latency, energy, and onboard processing concerns [29]. Pipeline B can use YOLO or another detector as a source, but the contribution begins after detection: entity encoding, georeferencing assumptions, live actor management, stale-state display, and operator use.

Vision-based UAV target geolocation has a long literature. Fixed-wing miniature air vehicle work localized ground targets from image pixels, pose, and camera geometry [30]. UAV video imagery studies have combined target geolocation, tracking, and filtering [31]. Vision-based tracking and motion estimation for moving targets has been demonstrated with UAV video and telemetry [32]. More recent work improves target geolocation using multi-view aerial images in long-range oblique photography [33]. These references make a strict point: Pipeline B must say “estimated georeferencing under calibrated camera and pose assumptions”, not “triangulation” or “ground-truth localization” unless [TBD: TBD-GEO] is completed.

UAV communication constraints are also well established. UAV wireless communications bring opportunities and link-design challenges [34]. Surveys identify connectivity, reliability, routing, security, and quality-of-service issues in UAV communication networks [35,36]. Multi-tier drone architectures for 5G/B5G networks further underline the role of aerial platforms in constrained and heterogeneous communication systems [37]. Pipeline B therefore cannot assume that video, semantic telemetry, command-and-control, and pose channels degrade identically. Robustness must be tested under explicit packet loss, jitter, interruption, and stale-state conditions.

3.5 Simulation, Runtime Engines, and Software Infrastructure

Unreal-based vehicle simulation and MAVLink-oriented tooling are not novel by themselves. AirSim provides high-fidelity visual and physical simulation for autonomous vehicles [38], and DroneVR demonstrates a web virtual-reality simulator for drone operators [39]. Pipeline B uses Unreal Engine and Cesium as rendering and geospatial infrastructure. Those components are implementation choices. Their use is relevant to reproducibility and HMD performance, but the paper’s scientific argument depends on whether dynamic semantic telemetry produces a useful, measurable operator display.

3.6 Positioning Statement

The paper’s novelty should be stated conservatively:

Pipeline B integrates detection-derived semantic telemetry, estimated georeferencing, Brain-mediated runtime state, Unreal/Cesium dynamic actor reconstruction, and HMD-facing VR interaction into a single evaluable UAV operator-display workflow.

The paper should avoid claiming that it substitutes for video, proves information-theoretic semantic optimality, guarantees safety for piloting, performs certified detect-and-avoid, or solves BVLOS regulation. Those claims are outside the available evidence and outside the proper scope of the architecture.

4. System Overview

4.1 Operator Scenario

The target scenario is a UAV mission in which a remote operator needs to monitor terrain, aircraft state, obstacles, people, vehicles, or mission objects while communication quality is constrained or intermittently degraded. The operator wears an HMD and can inspect a synthetic 3D scene containing terrain context, UAV pose, detected entities, freshness indicators, and uncertainty indicators.

The proposed display is auxiliary. It is not the primary command-and-control safety channel. It should be used alongside regulatory control links, video when available, map displays, flight telemetry, and operational procedures. The display is valuable only if it helps the operator understand spatial relationships and object persistence without hiding uncertainty.

4.2 Architecture

Pipeline B has five runtime layers:

1. **Detection layer:** a YOLO or other detection source observes image frames or sensor data and emits object detections.
2. **Georeferencing layer:** detections are associated with estimated world coordinates using camera, pose, attitude, and terrain assumptions.
3. **Brain layer:** the ground-station service ingests detections, normalizes entities, manages freshness, and publishes operator-display state.
4. **Rendering layer:** Unreal Engine, Cesium terrain, and a telemetry component reconstruct dynamic actors from the current entity layer.
5. **Operator layer:** the HMD presents an immersive scene with entity class, confidence, age, uncertainty, and stale-state encodings.

Figure 1 summarizes the proposed runtime flow. It is an architecture figure, not experimental evidence.

4.3 Implemented Runtime Path

The current implemented path is organized around two HTTP interfaces:

1. a detection source publishes objects to `POST /api/obstacles`;
2. Brain normalizes source identity, class, confidence, timestamp, and position;
3. Brain exposes the current operator-display state through `GET /api/ui/data`;
4. `UPorcelTelemetryComponent` in Unreal polls the state endpoint;
5. Unreal spawns, updates, marks stale, or despawns actors using `entity_id` or `object_id`;

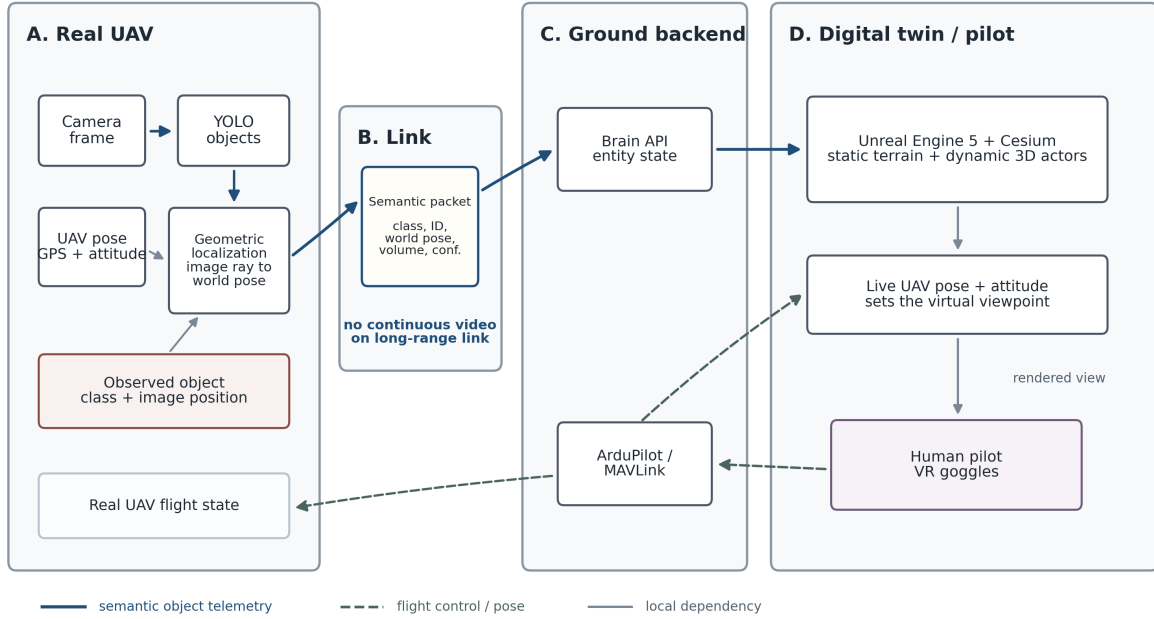


Figure 1 Pipeline B runtime flow. The UAV or upstream perception source detects objects with YOLO or another detector, estimates world coordinates under calibrated camera, pose, and terrain assumptions, and sends compact semantic telemetry to Brain. Brain publishes the live entity layer, while Unreal Engine, Cesium terrain, ArduPilot telemetry, and the live object layer are fused into an auxiliary VR operator display. Solid arrows denote semantic object telemetry, dashed arrows denote flight-control and pose feedback, and thin gray arrows denote local dependencies.

6. the VR-equipped operator observes the reconstructed entity layer in the digital twin.

Table 1 Pipeline B runtime data flow.

Stage	Producer	Consumer and required evidence
Object detection	YOLO or detection source	Telemetry encoder; detector model and confidence threshold [TBD: TBD-DEMO detector configuration]
Estimated georeferencing	UAV pose, camera model, terrain assumption	Detection message; calibration and uncertainty model [TBD: TBD-GEO calibration package]
Object ingest	Detection sender	Brain endpoint; request schema, timing logs, validation errors [TBD: TBD-LAT Brain ingest logs]
State publication	Brain	Unreal component; polling interval and JSON schema [TBD: TBD-REPRO API schema]
Actor reconstruction	Unreal component	VR operator display; spawn/update/despawn logs [TBD: TBD-DEMO actor lifecycle logs]
Immersive display	Unreal/Cesium runtime	HMD pilot view; screenshots, videos, headset runtime [TBD: TBD-DEMO HMD capture]

4.4 Entity Contract

Each reconstructed object is represented as a display entity:

$$e_i(t) = \{id_i, c_i, q_i, p_i, \Sigma_i, s_i, a_i, g_i, m_i\}, \quad (1)$$

where id_i is a stable identity, c_i is the class label, q_i is detector confidence, p_i is estimated position, Σ_i is position uncertainty, s_i is source identifier, a_i is age or freshness, g_i is optional geometry, and m_i is metadata for display and logging.

The minimum runtime fields are:

- `entity_id` or `object_id`;
- object class and optional subclass;
- confidence score and confidence-threshold policy;
- source frame timestamp and Brain receive timestamp;
- geospatial position `lat/lon/alt` when available;
- local coordinates `world_m` when available;
- approximate footprint, bounding volume, or class-specific geometry;
- source identifier and detector metadata;
- freshness, stale flag, and uncertainty fields [TBD: TBD-SAFETY freshness and uncertainty schema].

4.5 Actor Lifecycle

The display actor lifecycle is:

$$\text{new} \rightarrow \text{confirmed} \rightarrow \text{updated} \rightarrow \text{stale} \rightarrow \text{removed}. \quad (2)$$

An entity is marked stale when:

$$a_i(t) = t - t_i > \tau_{\text{stale}}, \quad (3)$$

where t_i is the last valid update time and τ_{stale} is the maximum age allowed before the display visually demotes the actor. The threshold must be selected from the operator task and verified experimentally [TBD: TBD-SAFETY max stale age]. An actor may be removed when:

$$a_i(t) > \tau_{\text{remove}}, \quad (4)$$

with $\tau_{\text{remove}} > \tau_{\text{stale}}$, unless the system policy requires persistent ghost actors for recent hazards [TBD: TBD-SAFETY despawn policy].

5. Method

5.1 Detection-to-World Projection

The georeferencing step must be framed as estimated world-coordinate assignment under calibrated camera and pose assumptions. For a pixel observation $z = [u, v]^T$, the normalized camera ray is:

$$\tilde{r}_c = K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad r_c = \frac{\tilde{r}_c}{\|\tilde{r}_c\|}, \quad (5)$$

where K is the camera intrinsic matrix. The corresponding world-frame ray is:

$$r_w = R_{wb} R_{bc} r_c, \quad (6)$$

where R_{bc} maps camera coordinates into the UAV body frame and R_{wb} maps the body frame into the world frame. Given UAV position p_u , an estimated object position is:

$$\hat{p}_o = p_u + \lambda r_w. \quad (7)$$

For a local planar ground assumption $z = z_g$, λ is:

$$\lambda = \frac{z_g - p_{u,z}}{r_{w,z}}. \quad (8)$$

For terrain intersection, z_g must be replaced by a terrain or map model [TBD: TBD-GEO terrain intersection model]. This projection is sensitive to camera calibration, UAV attitude, timestamp synchronization, terrain height, lens distortion, and the chosen pixel point for the detection. Therefore, the projected object position is an estimate, not a proof of target location.

5.2 Uncertainty Propagation

The display must avoid overconfidence. The uncertainty of the estimated object position can be approximated by first-order propagation:

$$\Sigma_p \approx J_x \Sigma_x J_x^T, \quad (9)$$

where x contains pixel coordinates, intrinsics, extrinsics, UAV pose, attitude, and terrain parameters, Σ_x is their covariance, and J_x is the Jacobian of the projection model with respect to those variables. The current draft does not yet have calibrated covariance values [TBD: TBD-GEO covariance inputs]. At minimum, the display must show a conservative uncertainty radius or confidence band [TBD: TBD-SAFETY uncertainty visualization].

5.3 Semantic Telemetry Bandwidth

For a mission interval of duration T , with N entity messages of serialized size $|m_k|$ bytes, semantic telemetry bitrate is:

$$B_{\text{sem}} = \frac{8}{T} \sum_{k=1}^N |m_k|. \quad (10)$$

Packet rate is:

$$R_{\text{pkt}} = \frac{N}{T}. \quad (11)$$

The corresponding video baseline must be measured over the same mission duration, scene, and link profile:

$$\Delta B = \frac{B_{\text{video}} - B_{\text{sem}}}{B_{\text{video}}}. \quad (12)$$

A bandwidth reduction is only meaningful if the semantic display preserves the entities needed for the operator task [TBD: TBD-BW entity completeness].

5.4 Latency Decomposition

End-to-end latency from source event to visible HMD update is:

$$L_{\text{e2e}} = t_{\text{hmd}} - t_{\text{det}}, \quad (13)$$

and should be decomposed as:

$$L_{\text{e2e}} = L_{\text{det}} + L_{\text{uplink}} + L_{\text{brain}} + L_{\text{poll}} + L_{\text{actor}} + L_{\text{render}} + L_{\text{hmd}}. \quad (14)$$

The software-only segments L_{brain} , L_{poll} , and L_{actor} can be instrumented immediately. The complete source-to-HMD measurement requires an HMD and a real or instrumented detection source [TBD: TBD-LAT e2e measurement].

5.5 VR Operator Display Requirements

The VRIH submission must describe the display as an immersive human-computer interface, not as a generic 3D plot. The paper therefore needs the following details:

- HMD model, runtime, refresh rate, resolution, tracking mode, and seated/standing configuration [TBD: TBD-DEMO HMD configuration];
- viewpoint modes: UAV-centric, external chase, top-down inspection, operator-selected orbit, or fixed mission overview [TBD: TBD-DEMO viewpoint modes];
- visual encodings for class, confidence, age, source, stale state, and uncertainty [TBD: TBD-SAFETY display encoding];
- interaction primitives: inspect entity, filter class, freeze history, recenter view, zoom, switch viewpoint, show raw source, and display uncertainty [TBD: TBD-HUMAN interaction model];
- fallback state when detections, pose, map, or network updates are stale [TBD: TBD-SAFETY fallback rule];
- comfort safeguards: frame rate, locomotion mode, camera motion, cybersickness screening, and session duration [TBD: TBD-HUMAN comfort protocol].

Figure 2 reserves the first missing HMD-facing figure.

[TBD: TBD-FIG-HMD: replace with HMD screenshot showing terrain, UAV pose, live actor, confidence, age, and uncertainty encoding.]

Figure 2 Placeholder for the pilot-facing VR digital-twin display. The final figure must be captured from the implemented HMD or HMD-mirrored runtime, not drawn as a mockup.

5.6 Software and Hardware Configuration

The reproducibility package must identify all software and hardware versions. The required configuration table is currently incomplete.

Table 2 Implementation configuration placeholders.		
Component	Current requirement	Status
Detection model	Architecture, weights, training domain, confidence threshold, class set	[TBD: TBD-REPRO YOLO/detector version and weights]
UAV platform	Airframe, autopilot, telemetry source, pose source, camera mount	[TBD: TBD-DEMO UAV or instrumented source]
Camera	Intrinsics, distortion, resolution, frame rate, exposure, calibration date	[TBD: TBD-GEO camera calibration]
Ground station	Brain host, OS, CPU/GPU, network adapter, clock sync	[TBD: TBD-REPRO Brain host configuration]
Unreal project	Unreal version, Cesium plugin version, project commit, map origin	[TBD: TBD-REPRO Unreal/Cesium versions]
VR headset	Device, runtime, refresh rate, tracking mode, capture method	[TBD: TBD-DEMO HMD configuration]
Network	Link type, bandwidth limit, packet-loss emulator, jitter profile	[TBD: TBD-LOSS network profile]

6. Current Readiness and Hardware Dependence

Software-only work can make the paper much stronger, but it cannot complete the VRIH research-paper claim. The submission-critical evidence still requires a real UAV or instrumented detection source, calibrated camera assumptions, an HMD, and human operators. Table 3 separates what can be done immediately from what is physically blocked.

Table 3 Work packages that can be completed now versus work requiring hardware.

Evidence item	Can be completed now	Requires hardware	Status
VR/H framing	Title, abstract, contribution, method, protocol, and limitations aligned to VR/HMD operator display.	No.	Done
Runtime contract	Document endpoints, schema, actor lifecycle, and display semantics.	No.	Drafted
API schema validation	Add JSON schema, accepted units, rejected fields, and log examples.	No.	[TBD: TBD- REPRO schema]
Bandwidth benchmark	Compute serialized message size, update rate, mean bitrate, p95 bitrate, and packet rate from replayed/synthetic streams.	Real mission logs improve validity.	[TBD: TBD-BW]
Brain-to-Unreal latency	Instrument sender, Brain ingest, state publication, Unreal polling, and actor update.	HMD not needed for partial segment.	[TBD: TBD-LAT partial]
Packet loss and jitter	Inject delay, loss, jitter, and interruption into semantic telemetry and observe stale behavior.	Real radio link optional for first test.	[TBD: TBD-LOSS]
Synthetic tracking	Replay known trajectories and measure ID switches, stale duration, fragmentation, and despawn.	No for synthetic validation.	[TBD: TBD- TRACK synthetic]
VR demonstration	Capture pilot-side HMD screenshots/video of the implemented prototype.	Yes: HMD and running Unreal VR configuration.	[TBD: TBD- DEMO]
Drone-to-VR latency	Measure source event to visible HMD update.	Yes: instrumented source plus HMD timing/capture.	[TBD: TBD-LAT e2e]
Geospatial error	Compare reconstructed positions against surveyed, RTK/GNSS, or motion-capture ground truth.	Yes: calibrated camera/pose and ground truth.	[TBD: TBD-GEO]
Human utility	Compare video-only, map-only, non-immersive hybrid, and VR digital twin conditions.	Yes: HMD, scenarios, participants, and ethics review/exemption.	[TBD: TBD- HUMAN]
Low visibility	Validate sensor behavior in night, fog, glare, smoke, or equivalent conditions.	Yes: sensor and environment-specific trials.	[TBD: TBD- LOWVIS]

7. Evaluation Protocol

7.1 Research Questions and Hypotheses

The evaluation should answer the following research questions:

- **RQ1:** Does semantic entity telemetry reduce bandwidth compared with video baselines for the same mission interval?
- **RQ2:** Is the source-to-HMD latency low enough for an auxiliary operator display?

- **RQ3:** Are estimated actor positions accurate enough, and is displayed uncertainty calibrated?
- **RQ4:** Does the display expose loss, jitter, interruption, stale detections, and tracking fragmentation without false freshness?
- **RQ5:** Does the VR display improve operator spatial awareness or task success relative to video-only, map-only, or non-immersive hybrid baselines?
- **RQ6:** Do low-visibility conditions change the system from useful to misleading when the sensor layer fails?

The corresponding hypotheses are:

- **H1:** semantic telemetry has lower mean and p95 bandwidth than video baselines while preserving required entity state [TBD: TBD-BW];
- **H2:** actor updates become visible in the HMD below a declared auxiliary-display latency threshold [TBD: TBD-LAT threshold];
- **H3:** geospatial error remains inside the displayed uncertainty envelope [TBD: TBD-GEO];
- **H4:** stale or uncertain actors are not perceived as fresh truth under link degradation [TBD: TBD-LOSS and TBD-HUMAN overtrust];
- **H5:** VR improves spatial awareness or task success without unacceptable workload or cybersickness [TBD: TBD-HUMAN];
- **H6:** any low-visibility benefit is conditional on validated sensor detections [TBD: TBD-LOWVIS].

7.2 Baselines

The minimum baselines are:

- **Video-only:** compressed camera feed with flight telemetry but no semantic reconstruction.
- **Map-only:** map or Cesium terrain with UAV pose but no dynamic semantic actors.
- **Non-immersive hybrid:** desktop or monitor view with map/video plus semantic overlays.
- **VR digital twin:** HMD display with terrain, UAV pose, dynamic actors, freshness, confidence, and uncertainty.

The final experiment must specify whether all baselines receive the same underlying detections, video source, network profile, and task instructions [TBD: TBD-HUMAN baseline parity].

7.3 Statistical Analysis Plan

The statistical plan must be written before collecting human-subject results. At minimum, the paper should report:

- sample size rationale and power or precision target [TBD: TBD-STAT sample size];
- within-subject or between-subject design, counterbalancing, and randomization [TBD: TBD-STAT design];
- primary endpoint, likely task success or spatial-awareness score [TBD: TBD-STAT primary endpoint];
- secondary endpoints: workload, near misses, overtrust, completion time, cybersickness [TBD: TBD-STAT secondary endpoints];
- model family, e.g., repeated-measures ANOVA, linear mixed model, generalized mixed model, or nonparametric alternative [TBD: TBD-STAT model];
- effect sizes and confidence intervals [TBD: TBD-STAT effect sizes];

- correction for multiple comparisons where needed [TBD: TBD-STAT multiplicity];
- exclusion criteria for simulator sickness, tracking failure, or incomplete trials [TBD: TBD-STAT exclusions].

This plan is intentionally included before results to prevent post-hoc metric selection.

8. Evaluation Results Pending Validation

This section reserves the exact structure of the final results. Each table is a placeholder and must be filled with measured values or replaced by a scoped limitation.

8.1 Real-System Demonstration

Table 4 Real-system demonstration placeholders.

Artifact	Purpose	Status	Acceptance criterion
HMD screenshots	Show terrain context, UAV pose, live actors, confidence, age, and uncertainty encoding.	[TBD: TBD-DEMO screenshots]	Captures come from the implemented HMD or HMD-mirrored runtime.
Operation video	Show spawn, update, stale marking, persistence, and despawn during operation.	[TBD: TBD-DEMO video]	Video links detections, telemetry, Brain state, and VR visualization.
Runtime logs	Support reproducibility of the demonstrated setup.	[TBD: TBD-DEMO logs]	Logs include endpoints, timestamps, HMD runtime, project commit, and display mode.
Supplementary stills	Provide VRIH video stills and graphical abstract candidates.	[TBD: TBD-FIG supplementary stills]	Still images are captured from real execution, not mockups.

8.2 Bandwidth Versus Video

Table 5 Bandwidth results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Mean bitrate	H.264, H.265, WebRTC FPV	[TBD: TBD-BW mean bitrate]	TBD	Lower bitrate for same scenario duration
P95 bitrate	Same codec and link profile	[TBD: TBD-BW p95 bitrate]	TBD	Reduced p95 load without entity loss
Packet rate	Same mission duration	[TBD: TBD-BW packet rate]	TBD	Compatible with target link budget
Entity completeness	Ground truth or replayed stream	[TBD: TBD-BW completeness]	TBD	Bandwidth reduction does not hide required state

8.3 End-to-End Latency

Table 6 Latency results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Detector to Brain	Synchronized detector timestamp and Brain receive log	[TBD: TBD-LAT detector-to-brain]	TBD	Below selected runtime threshold
Brain to Unreal actor	Polling and render logs	[TBD: TBD-LAT actor update]	TBD	P95 below operator-display threshold
Unreal to HMD display	Unreal timing and headset capture	[TBD: TBD-LAT vr-display]	TBD	Visible update within threshold
Source-to-HMD e2e	Detector event to actor visible in HMD	[TBD: TBD-LAT e2e-source-to-vr]	TBD	No unnoticed stale display

[TBD: TBD-FIG-LAT: replace with timestamp instrumentation diagram and latency waterfall.]

Figure 3 Placeholder for latency instrumentation. The final figure should show detector timestamp, Brain receive time, state publication, Unreal polling/update, render time, and HMD-visible update.

8.4 Geospatial Error

Table 7 Geospatial error results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
2D position error	RTK/GNSS or surveyed ground truth	[TBD: TBD-GEO 2D error]	TBD	Error inside declared uncertainty radius
3D position error	Ground-truth height or volume	[TBD: TBD-GEO 3D error]	TBD	Height error acceptable for visualization
Error by range	Near, mid, far detections	[TBD: TBD-GEO by range]	TBD	Error envelope reported by range bin
Uncertainty calibration	Measured error versus displayed uncertainty	[TBD: TBD-GEO uncertainty calibration]	TBD	Displayed uncertainty is not overconfident

8.5 Packet Loss and Jitter Robustness

Table 8 Packet-loss and jitter results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Packet loss degradation	Video-only and hybrid display	[TBD: TBD-LOSS loss sweep]	TBD	Graceful stale-state behavior
Jitter sensitivity	Fixed jitter profiles	[TBD: TBD-LOSS jitter sweep]	TBD	No false freshness under delayed updates
Interruption recovery	Link interruption scenario	[TBD: TBD-LOSS recovery]	TBD	Recovery state visible to operator
False freshness rate	Delayed or dropped updates	[TBD: TBD-LOSS false freshness]	TBD	Stale data is not presented as fresh

8.6 Dynamic Object Tracking

Table 9 Tracking results placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
ID switches	Ground-truth trajectories	[TBD: TBD-TRACK ID switches]	TBD	ID switches reported and bounded
Track fragmentation	Moving-object trial	[TBD: TBD-TRACK fragmentation]	TBD	Fragmentation does not hide uncertainty
Stale-object duration	Controlled occlusion or loss	[TBD: TBD-TRACK stale duration]	TBD	Stale state visibly encoded
Despawn correctness	Known disappearance events	[TBD: TBD-TRACK despawn]	TBD	Actors removed or demoted according to policy

8.7 Human Operator Utility

Table 10 Human-operator study placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
Task success	Video-only, map-only, hybrid	[TBD: TBD-HUMAN task success]	TBD	Improvement or non-inferiority with lower link load
Spatial awareness	SAGAT, SART, or equivalent	[TBD: TBD-HUMAN awareness]	TBD	Operators identify object position, age, and relevance
Workload	NASA-TLX or equivalent	[TBD: TBD-HUMAN workload]	TBD	Benefit without unacceptable cognitive load
Near misses	Same mission task	[TBD: TBD-HUMAN near misses]	TBD	No increase in safety-relevant events
Overtrust	Stale-state trials and questionnaire	[TBD: TBD-HUMAN overtrust]	TBD	Operators notice uncertainty and stale actors
Cybersickness	SSQ or equivalent	[TBD: TBD-HUMAN cybersickness]	TBD	No unacceptable sickness or dropout rate

8.8 Low-Visibility and Sensor Dependence

Table 11 Low-visibility validation placeholder.

Metric	Baseline	Pipeline B	Status	Acceptance criterion
RGB night or glare	Raw video and semantic display	[TBD: TBD-LOWVIS RGB]	TBD	Missed detections explicitly represented
Fog, smoke, or haze	Same mission with visibility degradation	[TBD: TBD-LOWVIS obscurant]	TBD	Sensor failure does not appear as clear scene
Sensor fallback	Thermal, LiDAR, radar, or equivalent	[TBD: TBD-LOWVIS sensor fallback]	TBD	Sensor limits reported by condition
Uncertainty display	Display variants	[TBD: TBD-LOWVIS uncertainty]	TBD	Operator can identify missing or stale state

9. Claim-to-Evidence Map

Table 12 Claims retained in the paper and evidence required before submission.

Claim	Required evidence	Status	Acceptance criterion
A VR digital-twin operator display has been implemented.	HMD screenshots, video, configuration records, and runtime logs showing detection-to-Brain-to-Unreal/VR operation.	[TBD: TBD-DEMO]	Demonstration material shows real execution, not only diagrams.
Semantic telemetry is hypothesized to reduce communication load relative to video.	Bitrate comparison against H.264, H.265, or WebRTC baselines for identical mission intervals.	[TBD: TBD-BW]	Semantic stream uses lower bandwidth while preserving task-relevant entities.
Actor updates are required to appear with acceptable delay.	Synchronized source, Brain, Unreal, and HMD timestamps.	[TBD: TBD-LAT]	P95 latency below declared auxiliary-display threshold.
Estimated actor positions are operationally useful.	Ground-truth comparison over range, altitude, class, terrain, and sensor conditions.	[TBD: TBD-GEO]	Error distribution inside declared uncertainty envelope.
The display exposes packet-loss degradation without false freshness.	Loss, jitter, and interruption experiments against video-only and hybrid baselines.	[TBD: TBD-LOSS]	Stale entities are visible as stale and never appear fresh.
Entity persistence is hypothesized to improve operator context.	Tracking metrics, stale-object duration, identity switches, fragmentation, and despawn correctness.	[TBD: TBD-TRACK]	Persistence helps memory without hiding uncertainty.
Human-operator benefit remains unvalidated.	Controlled study with task success, near misses, workload, spatial awareness, and overtrust.	[TBD: TBD-HUMAN]	Benefit is statistically or operationally meaningful.
Low-visibility operation is useful only with validated sensors.	Sensor-specific trials for RGB, thermal, LiDAR, radar, or equivalent inputs.	[TBD: TBD-LOWVIS]	Display distinguishes observed, inferred, stale, and missing objects.
The system is reproducible.	Code, schema, logs, versions, fixtures, and experimental scripts.	[TBD: TBD-REPRO]	Independent reader can replay or inspect the runtime path.

10. Limitations and Safety Envelope

Pipeline B can mislead an operator if uncertainty is hidden. The following limitations are part of the system definition:

- false negatives are critical because an absent actor may mean an undetected object, not a safe empty space;
- stale actors can look authoritative unless age, source, and confidence are visible;
- a visually stable synthetic scene can increase overtrust;
- object metadata, timestamps, and geolocation can leak privacy-sensitive information;
- low visibility, glare, occlusion, or smoke reduce sensor performance unless alternative sensors are validated;
- Cesium, map, terrain, geoid, origin, or coordinate-system errors can become actor-placement errors;
- timestamp drift can turn correct detections into incorrect moving-object states;
- VR can introduce cybersickness, attentional tunneling, fatigue, and display occlusion;
- BVLOS use requires detect-and-avoid, command-and-control reliability, operational design-domain limits, and regulatory compliance beyond this display.

The safety envelope must define:

- maximum stale-object age [TBD: TBD-SAFETY max stale age];
- maximum position uncertainty radius [TBD: TBD-SAFETY max uncertainty radius];
- minimum detection confidence for confirmed actor display [TBD: TBD-SAFETY minimum confidence];
- fallback rule when telemetry, detections, pose, or map context become stale [TBD: TBD-SAFETY fallback rule];
- allowed operating envelope by altitude, range, speed, terrain, lighting, link type, and sensor configuration [TBD: TBD-SAFETY allowed operating envelope];
- explicit prohibition against using the display as certified detect-and-avoid unless separate certification evidence exists [TBD: TBD-SAFETY regulatory wording].

11. Discussion

The central scientific risk is overclaiming. A reviewer can accept a paper that defines an honest VR operator-display architecture and validates it carefully. A reviewer should reject a paper that implies safe piloting, primary navigation, or BVLOS compliance from a visual prototype. Pipeline B is therefore framed as an auxiliary display for spatial awareness under degraded or constrained communication.

The second risk is treating a beautiful HMD scene as evidence. The scene only becomes evidence when it is tied to logs, timestamps, ground truth, and user-task outcomes. This is why the manuscript includes the claim-to-evidence map and placeholder result tables. Those tables are not decoration; they are the minimum conditions for the final paper.

The third risk is confusing semantic telemetry with semantic communication theory. Pipeline B transmits task-relevant entity state, but it does not currently prove semantic optimality, mutual information bounds, or learned communication objectives. The paper should therefore use “semantic telemetry” in an engineering sense and reserve stronger theoretical language for future work.

The fourth risk is human factors. VR may improve spatial awareness, but it may also increase workload, sickness, distraction, or overtrust. A VRIH submission must therefore include a real HMD task, baselines, subjective and objective measures, and a statistical plan.

12. Conclusion

Pipeline B is best framed for VRIH as a human-in-the-loop VR digital-twin operator display for dynamic UAV scene reconstruction from in-flight detections. The implemented runtime path converts detected objects into semantic telemetry, publishes a live entity layer through Brain, and reconstructs dynamic actors inside an Unreal/Cesium scene for HMD-based operator inspection.

The paper does not yet contain final experimental evidence. It now contains the full structure needed to reach a submission-ready version: state-of-the-art positioning, a conservative novelty claim, runtime contract, formal georeferencing and latency models, bandwidth and stale-state definitions, VR interaction requirements, evaluation protocol, placeholder result tables, safety envelope, statistical plan, and declarations. The remaining blockers are physical and experimental: HMD demonstration, real or instrumented source-to-HMD latency, calibrated geospatial validation, packet-loss trials, dynamic tracking validation, human-operator study, and low-visibility sensor evidence.

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CRedit Author Statement

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Declaration of Competing Interest

[TBD: Declaration of competing interests. If none, use the journal's declaration-tool wording and upload the required declaration document.]

Data and Code Availability

[TBD: Repository, commit hash, dataset/log availability, synthetic replay fixtures, JSON schemas, analysis scripts, video artifacts, HMD screenshots, and any restrictions caused by privacy, UAV flight rules, or third-party software.]

Ethics Statement

[TBD: Human-operator study ethics approval or exemption, informed consent procedure, participant inclusion/exclusion criteria, simulator-sickness handling, data anonymization, and withdrawal protocol.]

Declaration of Generative AI and AI-Assisted Technologies

[TBD: Final declaration required by VRIH/Elsevier if AI tools were used in manuscript preparation. The final authors must review, verify, and take responsibility for all content, references, and claims. AI tools cannot be listed as authors.]

Submission Materials Still Required

- final title page with complete affiliations, postal address, phone number, email, and ORCID where available [TBD: TBD-SUBMISSION title page];
- final abstract under 250 words and 1–7 keywords [TBD: TBD-SUBMISSION final abstract check];
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- video files or links plus still images if supplementary video is submitted [TBD: TBD-SUBMISSION video];
- editable LaTeX source, VRIH class file, figures, and any supplementary files [TBD: TBD-SUBMISSION editable source];
- reference audit confirming cited sources, DOI links, and citation-list consistency [TBD: TBD-SUBMISSION reference audit].

References

- 1 Steuer J. Defining virtual reality: Dimensions determining telepresence. *Journal of Communication*, 1992, 42(4): 73–93. doi: <https://doi.org/10.1111/j.1460-2466.1992.tb00812.x>
- 2 Sheridan T B. Musings on telepresence and virtual presence. *Presence: Teleoperators and Virtual Environments*, 1992, 1(1): 120–126. doi: <https://doi.org/10.1162/pres.1992.1.1.120>
- 3 Endsley M R. Toward a theory of situation awareness in dynamic systems. *Human Factors*, 1995, 37(1): 32–64. doi: <https://doi.org/10.1518/001872095779049543>
- 4 Endsley M R. Measurement of situation awareness in dynamic systems. *Human Factors*, 1995, 37(1): 65–84. doi: <https://doi.org/10.1518/001872095779049499>
- 5 Moens L L, Lydon S, Cucurachi S, et al. Measuring situation awareness: A meta-review across domains. *Human Factors*, 2026, 68(5): 632–672. doi: <https://doi.org/10.1177/00187208251412110>
- 6 Hart S G, Staveland L E. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In: Hancock P A, Meshkati N, eds. *Human Mental Workload*. Amsterdam: North-Holland, 1988. 139–183. doi: [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- 7 Hart S G. NASA-Task Load Index (NASA-TLX); 20 years later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 2006, 50(9): 904–908. doi: <https://doi.org/10.1177/154193120605000909>
- 8 Endsley M R, Selcon S J, Hardiman T D, et al. A comparative analysis of SAGAT and SART for evaluations of situation awareness. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 1998, 42(1): 82–86. doi: <https://doi.org/10.1177/154193129804200119>
- 9 Kennedy R S, Lane N E, Berbaum K S, et al. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The International Journal of Aviation Psychology*, 1993, 3(3): 203–220. doi: https://doi.org/10.1207/s15327108ijap0303_3
- 10 Fabris E J, Sangalli V A, Soares L P, et al. Immersive telepresence on the operation of unmanned vehicles. *International Journal of Advanced Robotic Systems*, 2021, 18(1). doi: <https://doi.org/10.1177/1729881420978544>
- 11 Roldán J, Peña-Tapia E, Martín-Barrio A, et al. Multi-robot interfaces and operator situational awareness: Study of the impact of immersion and prediction. *Sensors*, 2017, 17(8): 1720. doi: <https://doi.org/10.3390/s17081720>
- 12 Luo Y, Wang J, Pan Y, et al. Visual augmentation of live-streaming images in virtual reality to enhance teleoperation of unmanned ground vehicles. *Frontiers in Virtual Reality*, 2024, 5. doi: <https://doi.org/10.3389/frvir.2024.1230885>
- 13 Buchner S L, Rindfuss A, Wood J, et al. Impacts of 3D visualizations and virtual reality in display designs for remote monitoring of satellite operations. *Frontiers in Virtual Reality*, 2025, 6. doi: <https://doi.org/10.3389/frvir.2025.1487281>
- 14 Costa C, Gomes E, Rodrigues N, et al. Augmented reality mobile digital twin for unmanned aerial vehicle wildfire prevention. *Virtual Reality*, 2025, 29(2). doi: <https://doi.org/10.1007/s10055-025-01145-w>
- 15 Yang Z, Tomita K, Kamimura A. VR-based teleoperation of UAV–manipulator systems: From single-UAV control to dual-UAV cooperative manipulation. *Applied Sciences*, 2025, 15(20): 11086. doi: <https://doi.org/10.3390/app152011086>
- 16 Vona F, Amer M, Abdellatif O, et al. Unmanned aerial vehicles control in a digital twin: Exploring the effect of different points of view on user experience in virtual reality. *Proceedings of the 2025 31st ACM Symposium on Virtual Reality Software and Technology*, 2025: 1–11. doi: <https://doi.org/10.1145/3756884.3766010>
- 17 Lv Z, Marfia G, Poesi F, et al. Virtual-reality and intelligent hardware in digital twins. *Virtual Reality & Intelligent Hardware*, 2022, 4(4): ii–iv. doi: <https://doi.org/10.1016/j.vrih.2022.08.002>
- 18 Ahmed I, Ahmad M, Jeon G. Integrating digital twins and deep learning for medical image analysis in the era of COVID-19. *Virtual Reality & Intelligent Hardware*, 2022, 4(4): 292–305. doi: <https://doi.org/10.1016/j.vrih.2022.03.002>
- 19 Jones D, Snider C, Nassehi A, et al. Characterising the digital twin: A systematic literature review. *CIRP Journal of Manufacturing Science and Technology*, 2020, 29: 36–52. doi: <https://doi.org/10.1016/j.cirpj.2020.02.002>

- 20 Barricelli B R, Casiraghi E, Fogli D. A survey on digital twin: Definitions, characteristics, applications, and design implications. *IEEE Access*, 2019, 7: 167653–167671. doi: <https://doi.org/10.1109/ACCESS.2019.2953499>
- 21 Singh M, Fuenmayor E, Hinchey E, et al. Digital twin: Origin to future. *Applied System Innovation*, 2021, 4(2): 36. doi: <https://doi.org/10.3390/asi4020036>
- 22 Lv Z, Chen D, Feng H, et al. Beyond 5G for digital twins of UAVs. *Computer Networks*, 2021, 197: 108366. doi: <https://doi.org/10.1016/j.comnet.2021.108366>
- 23 Lei L, Shen G, Zhang L, et al. Toward intelligent cooperation of UAV swarms: When machine learning meets digital twin. *IEEE Network*, 2021, 35(1): 386–392. doi: <https://doi.org/10.1109/MNET.011.2000388>
- 24 Luo X, Chen H H, Guo Q. Semantic communications: Overview, open issues, and future research directions. *IEEE Wireless Communications*, 2022, 29(1): 210–219. doi: <https://doi.org/10.1109/MWC.101.2100269>
- 25 Xie H, Qin Z, Li G Y, et al. Deep learning enabled semantic communication systems. *IEEE Transactions on Signal Processing*, 2021, 69: 2663–2675. doi: <https://doi.org/10.1109/TSP.2021.3071210>
- 26 Shao J, Mao Y, Zhang J. Learning task-oriented communication for edge inference: An information bottleneck approach. *IEEE Journal on Selected Areas in Communications*, 2022, 40(1): 197–211. doi: <https://doi.org/10.1109/JSAC.2021.3126087>
- 27 Shao J, Zhang X, Zhang J. Task-oriented communication for edge video analytics. *IEEE Transactions on Wireless Communications*, 2024, 23(5): 4141–4154. doi: <https://doi.org/10.1109/TWC.2023.3314888>
- 28 Shao J, Mao Y, Zhang J. Task-oriented communication for multidevice cooperative edge inference. *IEEE Transactions on Wireless Communications*, 2023, 22(1): 73–87. doi: <https://doi.org/10.1109/TWC.2022.3191118>
- 29 Cao Z, Kooistra L, Wang W, et al. Real-time object detection based on UAV remote sensing: A systematic literature review. *Drones*, 2023, 7(10): 620. doi: <https://doi.org/10.3390/drones7100620>
- 30 Barber D B, Redding J D, McLain T W, et al. Vision-based target geo-location using a fixed-wing miniature air vehicle. *Journal of Intelligent and Robotic Systems*, 2006, 47(4): 361–382. doi: <https://doi.org/10.1007/s10846-006-9088-7>
- 31 Hosseinpoor H R, Samadzadegan F, Dadrasjavan F. Precise target geolocation and tracking based on UAV video imagery. *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 2016, XLI-B6: 243–249. doi: <https://doi.org/10.5194/isprsarchives-XLI-B6-243-2016>
- 32 Dobrokhodov V N, Kaminer I I, Jones K D, et al. Vision-based tracking and motion estimation for moving targets using unmanned air vehicles. *Journal of Guidance, Control, and Dynamics*, 2008, 31(4): 907–917. doi: <https://doi.org/10.2514/1.33206>
- 33 Liu C, Ding Y, Zhang H, et al. Improving target geolocation accuracy with multi-view aerial images in long-range oblique photography. *Drones*, 2024, 8(5): 177. doi: <https://doi.org/10.3390/drones8050177>
- 34 Zeng Y, Zhang R, Lim T J. Wireless communications with unmanned aerial vehicles: Opportunities and challenges. *IEEE Communications Magazine*, 2016, 54(5): 36–42. doi: <https://doi.org/10.1109/MCOM.2016.7470933>
- 35 Gupta L, Jain R, Vaszkun G. Survey of important issues in UAV communication networks. *IEEE Communications Surveys & Tutorials*, 2016, 18(2): 1123–1152. doi: <https://doi.org/10.1109/COMST.2015.2495297>
- 36 Hayat S, Yanmaz E, Muzaffar R. Survey on unmanned aerial vehicle networks for civil applications: A communications viewpoint. *IEEE Communications Surveys & Tutorials*, 2016, 18(4): 2624–2661. doi: <https://doi.org/10.1109/COMST.2016.2560343>
- 37 Sekander S, Tabassum H, Hossain E. Multi-tier drone architecture for 5G/B5G cellular networks: Challenges, trends, and prospects. *IEEE Communications Magazine*, 2018, 56(3): 96–103. doi: <https://doi.org/10.1109/MCOM.2018.1700666>
- 38 Shah S, Dey D, Lovett C, et al. AirSim: High-fidelity visual and physical simulation for autonomous vehicles. In: Hutter M, Siegwart R, eds. *Field and Service Robotics*. Cham: Springer, 2017. doi: https://doi.org/10.1007/978-3-319-67361-5_40
- 39 Nguyen V T, Jung K, Dang T. DroneVR: A web virtual reality simulator for drone operator. 2019 IEEE International Conference on Artificial Intelligence and Virtual Reality, 2019: 257–2575. doi: <https://doi.org/10.1109/AIVR46125.2019.00060>